KG COLLEGE OF ARTS AND SCIENCE
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JAVA PROJECT REPORT ON

PROJECT TITLE

Submitted by

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25BCS015

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DECLARATION

I hereby declare that the project entitled "**SALARY PAYROLL PROCESSING AND PAYSLIP GENERATION SYSTEM"** submitted in partial fulfillment of the requirements for the course **[JAVA PROGRAMMING LANGUAGE]** is a **record of my original work** carried out under the guidance of **[Dr.BOOPALAN SUNDARAMOORTHY]**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

DHAKSHANA K

(Your Name)

CERTIFICATE

This is to certify that the project entitled **"SALARY PAYROLL PROCESSING AND PAYSLIP GENERATION SYSTEM"** is a **bona fide work** carried out by **[DHAKSHANA K] [I-A/2025-2028]**, [**COMPUTER SCIENCE]**, **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **[JAVA PROGRAMMING LANGUAGE]**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: Coimbatore

Date: 10/04/2026

Submitted for the Viva-Voce Examination held on 10/04/2026

Guide:
(Technical Trainer, KGCAS)

Head of the Department
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ABSTRACT

The Salary Payroll Processing and Payslip Generation System is a Java-based application designed to automate employee salary management in organizations. In many small companies, payroll calculations are done manually, which can lead to errors, delays, and poor record management.

This system simplifies payroll processing by calculating employee salaries based on basic pay, allowances, and deductions such as tax, provident fund, and other benefits. It automatically generates a payslip that clearly displays salary details.

The application is developed using Core Java concepts such as classes, objects, methods, loops, and conditional statements. It can be implemented as a console-based application or using Java Swing for a graphical interface.

The main objective is to reduce manual work, improve accuracy, and ensure efficient salary processing. The system generates structured payslips including earnings, deductions, and net salary.

INTRODUCTION

Payroll management is an important task in every organization. Traditionally, many companies calculate employee salaries manually using spreadsheets or paper-based methods. This process is time-consuming and prone to errors.

With the advancement of technology, automated payroll systems have become essential. These systems help organizations manage employee salary details efficiently and generate accurate payslips.

The Salary Payroll Processing and Payslip Generation System is designed to automate salary calculations and generate payslips. The system allows the user to input employee details such as name, ID, basic salary, and allowances. It calculates deductions like tax and provident fund and generates the final salary.

This project uses Core Java concepts and demonstrates how programming can be applied in real-world business applications.

1.1 Objective

The main objectives are:

• To develop a Java-based payroll system

• To automate salary calculation

• To reduce manual errors

• To generate accurate payslips

• To implement Core Java concepts

• To provide a user-friendly system

1.2 Overview

The system is a menu-driven application where the user enters employee details and salary components.

It performs:

• Salary calculation

• Deduction calculation

• Net salary generation

• Payslip generation

The system simplifies payroll operations and improves efficiency.

1.3 Features

• Employee Details Entry

• Salary Calculation

• Deduction Calculation (Tax, PF)

• Payslip Generation

• Fast Processing

• User-Friendly Interfac

PROGRAM ANALYSIS

Program analysis is an important phase in software development where the problem is studied in detail and system requirements are identified. This phase helps in understanding how the system should function and what features are needed to solve the problem effectively.

In the case of the Salary Payroll Processing and Payslip Generation System, the analysis focuses on identifying the issues in the existing manual payroll system and proposing an automated solution to improve efficiency, accuracy, and record management.

2.1 Problem Definition

In many small organizations and businesses, employee salary processing is done manually using paper records or spreadsheets. This traditional method creates several challenges.

Some common problems include:

• Errors in salary calculation

• Time-consuming payroll processing

• Difficulty in managing multiple employee records

• Incorrect deduction calculations (tax, PF, etc.)

• Lack of proper payslip generation

• Poor data management and record keeping

Manual systems increase workload and may lead to employee dissatisfaction due to incorrect salary or delays.

Therefore, there is a need for an automated payroll system that can efficiently calculate salaries, manage employee data, and generate accurate payslips.

The Salary Payroll Processing and Payslip Generation System is designed to solve these issues by providing a computerized solution.

2.2 Project Overview

The Salary Payroll Processing and Payslip Generation System is a Java-based application developed to automate salary processing in an organization.

The system provides a simple interface where users can:

• Enter employee details (Name, ID, Basic Salary)

• Calculate allowances such as HRA and DA

• Calculate deductions such as Tax and Provident Fund

• Generate net salary automatically

• Produce a structured payslip

The system performs all calculations automatically and displays the final salary details clearly.

This project is implemented using Core Java programming concepts and can be developed as a console-based application or with a graphical interface using Java Swing.

2.3 Scopes and Limitations

Scope

The scope of the system includes:

• Managing employee salary details

• Automating payroll calculations

• Generating employee payslips

• Reducing manual effort and errors

• Demonstrating Core Java concepts

The system can be used in:

• Small companies

• Offices

• Educational institutions

• Shops and small businesses

Limitations

• Does not store data permanently (without database integration)

• Limited to basic payroll calculations

• Not suitable for large-scale enterprise use

• No advanced features like online payment or cloud storage

SYSTEM DESIGN

System design is the stage where the structure and components of the system are planned before implementation. In this phase, the system is divided into different modules, and the workflow is clearly defined. A good system design makes the program easy to develop, understand, and maintain.

The Salary Payroll Processing and Payslip Generation System is designed as a user-friendly application where users can enter employee details, calculate salary components, and generate payslips automatically.

The system is divided into several modules, each responsible for a specific function.

3.1 Modules

The main modules of the **Salary Payroll Processing and Payslip Generation System** are explained below:

The main modules of the Salary Payroll Processing and Payslip Generation System are explained below:

1. Employee Details Module

This module is responsible for collecting and storing employee information.

Functions of this module include:

• Accepting employee name

• Accepting employee ID

• Accepting basic salary

Example:

Employee Name: John

Employee ID: 101

Basic Salary: ₹20,000

This module acts as the input stage for the system.

2. Salary Calculation Module

This module calculates the earnings of the employee based on the basic salary.

The system calculates:

• HRA (House Rent Allowance) – e.g., 20% of basic salary

• DA (Dearness Allowance) – e.g., 10% of basic salary

• Gross Salary = Basic Salary + HRA + DA

Example Calculation:

HRA = 20% of 20,000 = 4,000

DA = 10% of 20,000 = 2,000

Gross Salary = 26,000

3. Deduction Module

This module calculates the deductions from the salary.

Deductions include:

• Tax (e.g., 5% of gross salary)

• Provident Fund (PF) (e.g., 2% of gross salary)

Example:

Tax = 5% of 26,000 = 1,300

PF = 2% of 26,000 = 520

Total Deduction = 1,820

4. Net Salary Calculation Module

This module calculates the final salary after deductions.

Formula:

Net Salary = Gross Salary – Total Deductions

Example:

Net Salary = 26,000 – 1,820 = 24,180

5. Payslip Generation Module

This module generates the final payslip for the employee.

The payslip includes:

• Employee Name

• Employee ID

• Basic Salary

• Allowances (HRA, DA)

• Gross Salary

• Deductions (Tax, PF)

• Net Salary

Example Output:

------ PAYSLIP ------

Employee Name: John

Employee ID: 101

Basic Salary: 20000

HRA: 4000

DA: 2000

Gross Salary: 26000

Tax: 1300

PF: 520

Net Salary: 24180

---

6. Exit Module

This module allows the user to safely exit the application.

When selected, the system terminates the program.

METHODOLOGY

Methodology describes the process followed to develop the project. It explains the steps taken from understanding the problem to designing, implementing, testing, and documenting the system.

For the Salary Payroll Processing and Payslip Generation System, a structured development approach was followed to ensure that the system meets requirements and works efficiently.

The development process includes the following stages:

• Problem Analysis

• System Design

• Program Development

• Testing and Validation

• Result and Documentation

4.1. Problem Analysis

In this phase, the existing manual payroll system is studied and analyzed in detail.

The main problems identified are:

• Manual salary calculation may lead to errors

• Payroll processing takes more time

• Difficult to manage multiple employees

• Incorrect deductions and calculations

• Lack of proper payslip generation

Based on these problems, the system requirements were defined to develop an automated payroll system that simplifies salary processing and improves accuracy.

4.2. System Design

In this stage, the structure of the system is planned before coding begins.

The system is divided into modules such as:

• Employee Details Module

• Salary Calculation Module

• Deduction Module

• Net Salary Module

• Payslip Generation Module

• Exit Module

The design ensures that users can easily interact with the system and generate payslips efficiently.

4.3. Program Development

After designing the system, the program is implemented using Java programming language.

The following steps were followed:

• Creation of classes and objects

• Writing methods for salary calculations

• Implementing logic for allowances and deductions

• Accepting user input

• Displaying the final payslip

Core Java concepts such as loops, conditional statements, methods, and object-oriented programming were used.

4.4. Testing and Validation

Testing ensures that the program works correctly and produces the expected results.

Different test cases were used to check the functionality of the system.

Example Test Cases:

Test Case	Expected Result
Enter employee details	Accepted correctly
Calculate salary	Correct output
Apply deductions	Accurate deduction
Generate payslip	Proper display
Invalid input	Error handled

This phase ensures the system is reliable and error-free.

4.5. Result and Documentation

After successful testing, the results are documented.

The system generates:

• Accurate salary calculations

• Proper deduction details

• Clear payslip output

Screenshots and outputs are included to demonstrate system functionality.

4.6. Algorithm

Algorithm for Salary Payroll Processing System:

1. Start the program

2. Enter employee details

3. Input basic salary

4. Calculate HRA and DA

5. Calculate gross salary

6. Calculate tax and PF

7. Calculate total deductions

8. Compute net salary

9. Display payslip

10. End the program

4.7. Flowchart

Students should draw a flowchart showing program flow.

Example flow:

Start
↓
Enter Employee Details
↓
Calculate Salary
↓
Apply Deductions
↓
Generate Payslip
↓
End

Students can draw using:

- draw.io
- Lucidchart
- MS Visio

SOURCE CODE

Include important parts of the Java code such as:

- Main class

```
public class Main {
        public static void main(String[] args) {
                PayrollSystem obj = new PayrollSystem();
                obj.generatePayslip("John", 101, 20000)
         }
}
```

- Menu logic

```
public class Main {
   public static void main(String[] args) {

      PayrollSystem obj = new PayrollSystem();
      obj.generatePayslip("John", 101, 20000);

   }
}
```

- Salary Calculation

```
double hra = basic * 0.20;
double da = basic * 0.10;

double gross = basic + hra + da;

double tax = gross * 0.05;
double pf = gross * 0.02;

double netSalary = gross - (tax + pf);
```

- Payslip Generation Method

```
public void generatePayslip(String name, int id, double basic) {

    double hra = basic * 0.20;
  double da = basic * 0.10;
    double gross = basic + hra + da;

    double tax = gross * 0.05;
    double pf = gross * 0.02;

    double netSalary = gross - (tax + pf);

    System.out.println("----- PAYSLIP -----");
    System.out.println("Name: " + name);
    System.out.println("ID: " + id);
    System.out.println("Net Salary: " + netSalary);
}
```

INPUT AND OUTPUT SCREENS

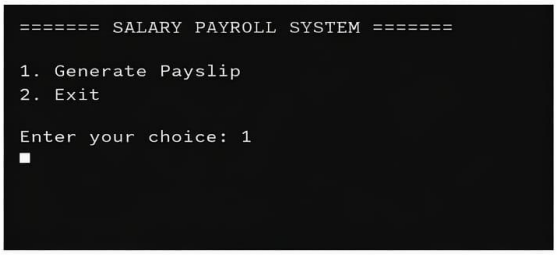


Figure 1: Program Menu

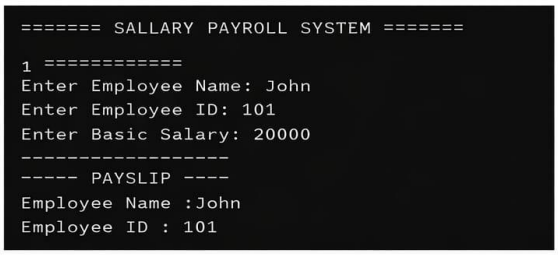


Figure 2: Item Selection

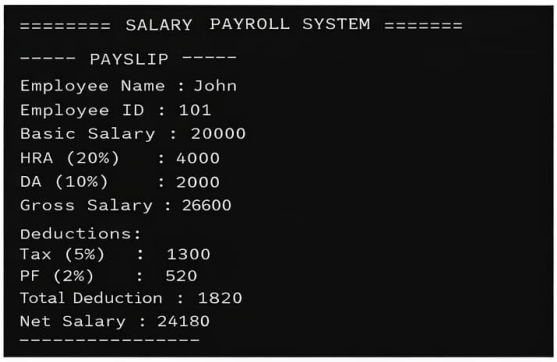


Figure 3: Billing Output

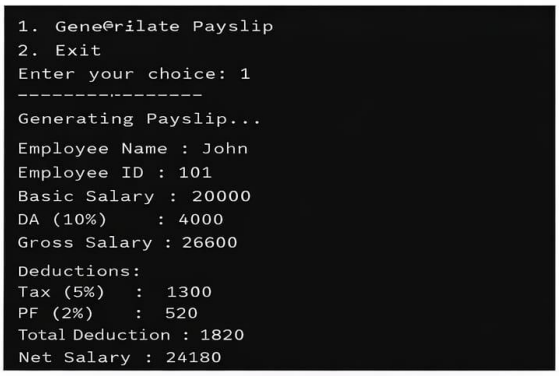


Figure 4: Final Invoice

DISCUSSION & CONCLUSION

7.1 Discussion

The Salary Payroll Processing and Payslip Generation System was developed to automate the process of employee salary calculation and payslip generation. The system allows users to enter employee details, calculate salary components such as allowances and deductions, and generate a structured payslip.

During the development of this project, Core Java programming concepts such as classes, objects, methods, loops, and conditional statements were applied effectively. The system performs accurate calculations for gross salary, deductions like tax and provident fund, and net salary.

The application was tested with different inputs to ensure correctness and reliability. The results show that the system reduces manual effort, minimizes calculation errors, and improves efficiency in payroll processing.

The system provides a simple and user-friendly interface, making it easy for users to operate even with basic computer knowledge.

7.2 Conclusion

The Salary Payroll Processing and Payslip Generation System successfully demonstrates the implementation of Core Java concepts in developing a real-world application.

The system automates salary processing, reduces human errors, and generates accurate payslips efficiently. It simplifies payroll management and saves time compared to manual methods.

This project helps in understanding how programming can be used to solve practical problems in organizations. It can be further enhanced by adding features such as database integration, graphical user interface (GUI), multi-user access, and online payroll management.

Overall, the project achieves its objective of creating a simple, efficient, and reliable payroll system.

REFERENCES

The following resources were used during the development of this project:

Books

• Herbert Schildt – *Java: The Complete Reference*

• *E. Balagurusamy – Programming with Java*

• *Y. Daniel Liang – Introduction to Java Programming*

Websites

• https://www.oracle.com/java

• https://www.geeksforgeeks.org

• https://www.w3schools.com

• https://www.javatpoint.com

• https://stackoverflow.com